

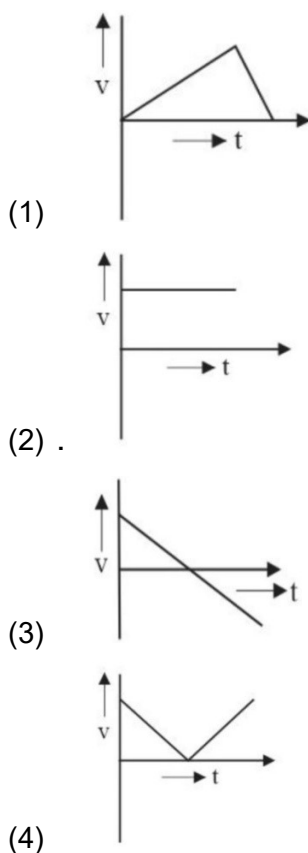
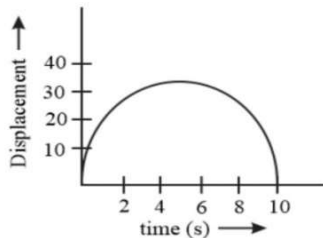


Accuracy Test # 1

Based on NTA Abhyas (PRE-MEDICAL)

KINEMATICS

1. The displacement-time graph of a moving object is shown in figure. Which of the velocity-time graphs shown in the figure could represent the motion of the same body?



2. The water drops falls at a regular intervals from a tap 5m above the ground. The third drop is leaving the tap at an instant when the first drop touches the ground. How far above the ground is the second drop at that instant? (take $g = 10 \text{ m/s}^2$)

- (1) 1.25 m
(2) 2.50 m
(3) 3.75 m
(4) 5.00 m

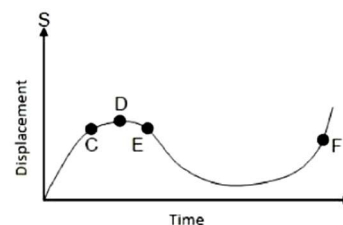
3. A projectile is fired with a velocity u at an angle θ with the ground. At some instant during motion, its velocity makes an angle α with the horizontal direction. The speed of the particle at this time will be-

- (1) $u \cos \theta \sec \alpha$
(2) $u \cos \theta \cos \alpha$
(3) $u^2 \cos^2 \alpha \sin^2 \alpha$
(4) $u \sin \theta \sin \alpha$

4. A ball rolls of top of the stair way with a horizontal velocity of the magnitude 1.8 m/s. The steps are 0.20m high and 0.20 m wide. Which step will be the ball hit first?

- (1) first
(2) second
(3) Third
(4) Fourth

5. The displacement time graph of a moving particle is shown below. The instantaneous velocity of the particle is negative at the point

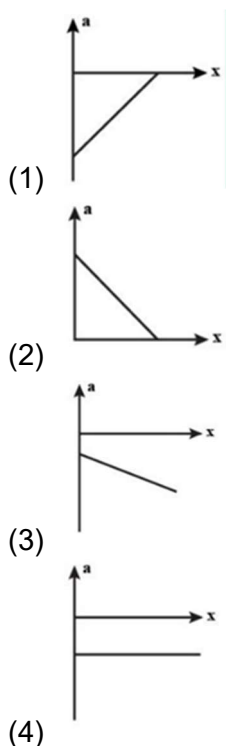
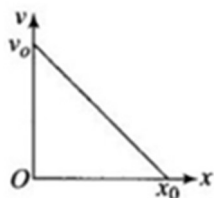


- (1) C
(2) E
(3) D
(4) F

6. A stone is thrown at 25m/s at 53° above the horizontal. At what time its velocity is at 45° below the horizontal?

- (1) 0.5s
(2) 4s
(3) 3.5 s
(4) 2.5 s

7. The velocity displacement graph of a particle moving along a straight line as shown in figure. The most suitable acceleration displacement graph will be-



8. A stone is dropped from a height of 45 m on a horizontal level ground. There is horizontal wind blowing due to which horizontal acceleration of the stone becomes 10 m/s^2 . (Take $g = 10 \text{ m/s}^2$). The time taken (t) by stone to reach the ground and the net horizontal displacement (x) of the stone from the time it is dropped and till it reaches the ground are respectively-

- (1) $t=3\text{s}$, $x=45\text{m}$
- (2) $t=4\text{s}$, $x=54\text{m}$
- (3) $t=3\text{s}$, $x=54\text{m}$
- (4) $t=4\text{s}$, $x=45\text{m}$

9. A ball is thrown upwards, it takes 4s to reach back to the ground. Find its initial velocity

- (1) 30 m/s
- (2) 100 m/s
- (3) 40 m/s
- (4) 20 m/s

10. If a ball is thrown vertically upwards with a speed u , the distance covered by it during the last t seconds of its ascent is:-

- 1) ut
- 2) $\frac{1}{2}gt^2$
- 3) $ut - \frac{1}{2}gt^2$
- 4) $(u+gt) t$

11. A particle is projected from ground at some angle with horizontal (Assuming point of projection to be origin and the horizontal and vertical directions to be the x and y - axis) the particle passes through the points (3, 4) m and (4, 3) m during its motion then the range of the particle would be: ($g = 10 \text{ m/s}^2$)

- (1) 36 m
- (2) $\frac{37}{7}$ m
- (3) 4m
- (4) 16 m

12. A ball is released from the top of a tower of height h meter. It takes T seconds to reach the ground. What is the position of the ball in $\frac{T}{3}$ second?

- (1) $\frac{17h}{18}$ metre from the ground
- (2) $\frac{8h}{9}$ metre from the ground
- (3) $\frac{7h}{9}$ metre from the ground
- (4) $\frac{h}{9}$ metre from the ground

13. Assertion: Average speed is equal to the magnitude of average velocity.

Reason: Displacement of body is less than or equal to distance.

- (1) Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of the Assertion.
- (3) Assertion is true but the Reason is false.
- (4) Assertion is false but the Reason is true.

14. The displacement x of a particle varies with time t as $x = a e^{-\alpha t} + b e^{\beta t}$, where a, b, α and β are positive constant. The velocity of the particle will:-

- (1) Be independent of β
- (2) Drop to zero, when $\alpha = \beta$
- (3) Decrease with time
- (4) Increase with time

15. A ball is projected from the bottom of an inclined plane of inclination 30° , with a velocity of 30 ms^{-1} at an angle of 30° with the inclined plane. If $g = 10 \text{ ms}^{-2}$, then the range of the ball on given inclined plane is:-

- (1) 12m
- (2) 60m
- (3) 120m
- (4) 600m

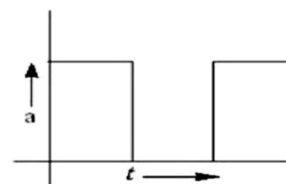
16. A projectile is thrown with velocity $u = 20 \text{ m/s} \pm 5\%$ at an angle 60° . If the projectile comes back on the ground at the same level then which of following cannot be a possible answer for range. (Take, $g = 10 \text{ m/s}^2$)

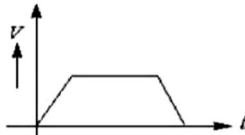
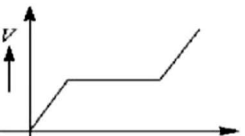
- (1) 34.6m
- (2) 37.5m
- (3) 32.0m
- (4) 39.0m

17. Two particles are projected simultaneously in the same vertical lane, from the same point, but with the different speeds and at different angles with the horizontal. The path followed by one, as seen by the other is-

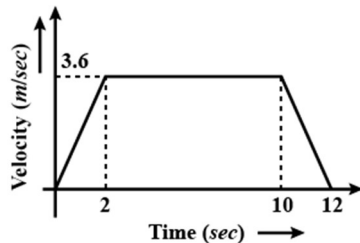
- (1) a vertical line
- (2) a parabola
- (3) a hyperbola
- (4) a straight line making a constant angle with the horizontal

18. Figure shows the acceleration time graph of a particle which of the following represents the corresponding velocity – time graph ?



- (1) 
- (2) 
- (3) 
- (4) 

- 19.** An elevator is going upward and the variation in the velocity of the elevator is as given in the graph. What is the height to which the elevator takes the passengers?



- (1) 3.6 m
(2) 28.8 m
(3) 36.0 m
(4) 32.4 m
- 20.** Three particles start from the origin at the same time, one with a velocity V_1 along the x-axis, second along the negative y-axis with a velocity V_2 and third particle moves along the line $x=y$. The velocity of third particle, so that three may always lie on the same line is
- (1) $\frac{v_1+v_2}{2}$
(2) $\sqrt{v_1 + v_2}$
(3) $\frac{v_1 v_2}{v_1 + v_2}$
(4) $v = \frac{\sqrt{2} v_1 v_2}{v_2 - v_1}$
- 21.** For an object thrown at 45° to horizontal, the maximum height H and horizontal range R are related as
- (1) $R = 16H$
(2) $R = 8H$
(3) $R = 4H$
(4) $R = 2H$

- 22.** A man running at a speed of 5km/h finds that the rain is falling vertically. When he stops running, he finds that the rain is falling at an angle of 60° with the horizontal. The velocity of rain with respect to running man is

- (1) $\frac{5}{\sqrt{3}}$ km/h
(2) $\frac{5\sqrt{3}}{2}$ km/h
(3) $\frac{4\sqrt{3}}{5}$ km/h
(4) $5\sqrt{3}$ km/h

- 23.** A particle is projected with velocity v_0 along x axis. The deceleration on the particle is proportional to the square of the distance from the origin i.e. $a = -x^2$. The distance at which article stops is

- (1) $\sqrt{\frac{3v_0}{2}}$
(2) $\left(\frac{v_0^2}{3}\right)^{\frac{1}{3}}$
(3) $\left(\frac{2v_0^2}{3}\right)^{\frac{1}{3}}$
(4) $\left(\frac{3v_0^2}{2}\right)^{\frac{1}{3}}$

- 24.** A ball is projected at an angle 60° with the horizontal with speed 30 m/s. what will be the speed of the ball when it makes an angle 45° with the horizontal ?

- (1) 30 m/s
(2) $15\sqrt{2}$ m/s
(3) 15 m/s
(4) $30\sqrt{2}$ m/s

25. Two bodies begin a free fall from rest from the same height 2 seconds apart. How long after the first body begins to fall, the two bodies will be 40 m apart? (take $g = 10\text{m/s}^2$)

- (1) 1 s
- (2) 2 s
- (3) 3 s
- (4) 4 s

26. A particle A is projected from the ground with an initial velocity of 10 m/s at an angle of 60° with horizontal. From what height h should another particle B be projected horizontally with velocity 5 m/s so that both the particles collide in ground at point C if both are projected simultaneously? ($g = 10\text{ms}^{-2}$)

- (1) 10 m
- (2) 30 m
- (3) 15 m
- (4) 25 m

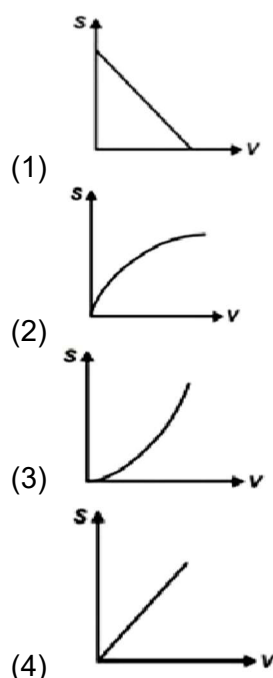
27. A particle is projected vertically upwards with average velocity u and from a point O. When it returns to the point of projection, which of the following is incorrect?

- (1) average velocity is zero
- (2) displacement is zero
- (3) average speed is u
- (4) average speed is $\frac{u}{2}$

28. Four persons K, L, M, N are initially at four corners of a square of side d . Each person now moves with a uniform speed v in such a way that K always moves directly towards L , L directly towards M , M directly towards N , N directly towards K . The four persons will meet at a time

- (1) $\frac{d}{v}$
- (2) $\frac{d}{2v}$
- (3) $\frac{2d}{v}$
- (4) The four persons will never meet

29. The displacement (s) – velocity (v) graph of a particle if it starts moving from rest with a uniform acceleration which is parallel to its instantaneous direction of motion is



30. A very broad elevator is going vertically upwards with a constant acceleration of 2ms^{-2} . At the instant when its velocity is 4ms^{-1} , a ball is projected from the floor of the lift with a speed of 4ms^{-1} relative to the floor at an angle 30° . The time of flight of the ball in the elevator is ($g = 10\text{ms}^{-2}$)

- (1) $\frac{1}{2}s$
- (2) $\frac{1}{3}s$
- (3) $\frac{1}{4}s$
- (4) 1s

31. Find the time after which the particle's initial velocity will be perpendicular to instantaneous velocity when it is projected with 30 m/s from horizontal ground by making an angle 60° with vertical.

- (1) 6 second
- (2) $2\sqrt{3}$ second
- (3) 3 second
- (4) Never possible

32. In ground to ground projectile motion under gravity, which of the following doesn't affect the time of flight?

- (1) Rotation of earth
- (2) Air resistance
- (3) Curvature of earth
- (4) All of these

33. Two particles are projected from a point at the same instant with velocities whose horizontal components and vertical components are (u_1, v_1) and (u_2, v_2) respectively. The time interval between their passing through the other common point of their path (other than origin) is

- (1) $\frac{2}{g} \left(\frac{v_1 u_1 - v_2 u_2}{u_1 + u_2} \right)$
- (2) $\frac{2}{g} \left(\frac{v_1^2 + v_2^2}{u_1 + u_2} \right)$
- (3) $\frac{2}{g} \left(\frac{u_1^2 + u_2^2}{v_1 + v_2} \right)$
- (4) $\frac{2}{g} \left(\frac{v_1 u_2 - v_2 u_1}{u_1 + u_2} \right)$

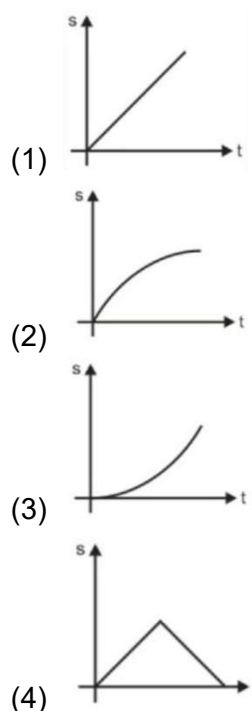
34. If a stone is to hit at a point which is at a horizontal distance 100 m away and at a height 50 m above the point from where the stone starts, then what is the value of initial speed u if the stone is launched at an angle 45° ?

- (1) $10\sqrt{2} \frac{m}{s}$
- (2) $10\sqrt{5} \frac{m}{s}$
- (3) $20\sqrt{5} \frac{m}{s}$
- (4) $20\sqrt{10} \frac{m}{s}$

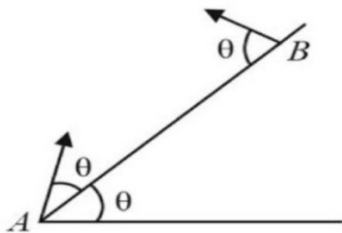
35. From the top of a tower, a stone is thrown up and reaches the ground in time $t_1 = 9$ s. a second stone is thrown down with the same speed and reaches the ground in time $t_2 = 4$ s. a third stone is released from rest and reaches the ground in time t_3 , which is equal to-

- (1) 6.5 s
- (2) 6.0 s
- (3) $72/13$ s
- (4) None

36. One stone is dropped from a tower from rest and simultaneously another stone is projected vertically upward from the tower with some initial velocity. The graph of the distance (s) between the two stones varies with time (t) as (before either stone hits the ground).



- 37.** From an inclined plane, two particles are projected with the same speed at same angle θ , one up and other down the plane as shown in the figure. Which of the following statement (s) is/are correct ?



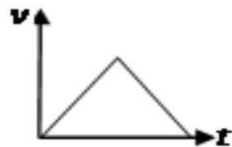
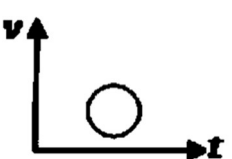
- (1) The time of flight of each particle is the same
 (2) The particles will collide the plane with same speed
 (3) Both the particles strike the plane perpendicularly
 (4) The particles will collide in mid air if projected simultaneously and time of flight of each particle is less than the time of collision.
- 38.** Check up the only correct statement in the following-
- (1) a body has a constant velocity and still it can have a varying speed
 (2) a body has a constant speed and but it can have a varying velocity
 (3) a body having constant speed cannot have any acceleration
 (4) a body in motion under a force acting upon must always have work done upon it
- 39.** A particle is thrown over a triangle from one end of a horizontal base and after grazing the vertex falls on the other end of the base. If 30° and 60° be the base angles and θ the angle of projection then $\tan \theta$ is-

- (1) $\frac{2}{\sqrt{3}}$
 (2) $\frac{4}{\sqrt{3}}$
 (3) $\frac{1}{3}$
 (4) 3

- 40.** Horizontal wind is blowing with a velocity v towards north-east. A man starts running towards north with acceleration a . The time after which man will feel the wind blowing towards east is

- (1) $\frac{v}{a}$
 (2) $\frac{v\sqrt{2}}{a}$
 (3) $\frac{v}{a\sqrt{2}}$
 (4) $\frac{2v}{a}$

- 41.** Which of the following curve does not represent motion of a body in a straight line ?

- (1) 
- (2) 
- (3) 
- (4) 

42. A projectile is thrown with a velocity of 20 m/s at an angle of 60° with the horizontal. After how much time the velocity vector will make an angle of 45° with the horizontal-

- (1) $\sqrt{3}$ s
- (2) $\frac{1}{\sqrt{3}}$ s
- (3) $(\sqrt{3} + 1)$ s
- (4) $(\sqrt{3} - 1)$ s

43. A particle is projected with a speed $10\sqrt{2}$ m/s and at an angle 45° with the horizontal. The rate of change of speed with respect to time at $t = 1$ s is ($g = 10 \text{ m/s}^2$)

- (1) $\frac{10}{\sqrt{2}}$ m/s²
- (2) 10 m/s^2
- (3) zero
- (4) 5 m/s^2

44. A particle is projected at an angle of 60° above the horizontal with a speed of 10 m/s. After some time the direction of its velocity makes an angle of 30° above the horizontal. The speed of the particle at this instant is

- (1) $\frac{5}{\sqrt{3}}$ m/s
- (2) $5\sqrt{3}$ m/s
- (3) 5 m/s
- (4) $\frac{10}{\sqrt{3}}$ m/s

45. If a body is projected with an angle θ to the horizontal, then

- (1) its velocity is always perpendicular to its acceleration
- (2) its velocity becomes zero at its maximum height
- (3) its velocity makes zero angle with the horizontal at its maximum height
- (4) just before hitting the ground, the direction of velocity coincides with the acceleration